KARIVENA VINAY KUMAR

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Education

- **RGM College of Engineering & Technology** — Andhra Pradesh, India
  Bachelor of Technology in Electronics and Communication Engineering; GPA: 7.27 — *July 2019 - June 2023*
  Courses: *Analog/Digital Electronics, Circuit Theory, Signals & Systems, VLSI Design, Embedded Systems, Communication Systems, IoT and Programming*

Skills Summary

- **Languages**: Python, SQL, Bash
- **Frameworks**: Apache Spark SonwSql
- **Tools**: Docker, Git, PostgreSQL, MySQL, Postman, GitHub Actions, Kafka
- **Platforms**: Linux, AWS, Databricks, Snowflake, IICS
- **Soft Skills**: Leadership, Event Management, Public Speaking

Experience

- **Cognizant** — Remote
  Data Engineer (Full-time) — *Sep 2023 – Present*
  - : Designed and developed scalable ETL pipelines using Informatica Intelligent Cloud Services (IICS) to ingest data from 10+ heterogeneous sources (CSV, APIs, RDBMS) into Snowflake Data Warehouse.
  - : Optimized complex SQL transformations and implemented data validation checks in Snowflake, improving query performance by 30% and ensuring high data integrity.
  - : Automated Linux-based health check scripts and monitoring workflows, reducing manual validation effort by 40% and enabling real-time reporting dashboards for stakeholders.
  - : Collaborated with cross-functional teams to build CI/CD pipelines for data workflows across AWS and Azure environments.
- **Cognizant** — Remote
  AIA AWS Intern (Full-time) — *Jul 2023 – Sep 2023*
  - : Designed and implemented distributed ETL workflows using Hadoop and Apache Spark to process large-scale structured and unstructured datasets.
  - : Built fault-tolerant data pipelines on AWS leveraging Databricks and cloud storage services, improving data processing efficiency and scalability.
  - : Optimized extraction, transformation, and loading (ETL) operations, reducing data processing time by 25% through Spark performance tuning and partition optimization.
  - : Worked with NoSQL databases including MongoDB and HBase to support real-time analytics and business decision-making use cases.
- **Cognizant** — Bangalore, India
  IoT AWS Intern — *Mar 2023 – Jul 2023*
  - : Developed Java-based backend applications using Spring Boot to support IoT device communication and real-time data processing.
  - : Designed and optimized MySQL and PostgreSQL databases, implementing indexing and query tuning techniques to improve performance and ensure data integrity.
  - : Architected and deployed an AWS-based IoT solution integrating smart home devices using AWS IoT Core, enabling secure device communication and cloud-based automation workflows.
  - : Configured cloud infrastructure services (EC2, S3, IAM) to support scalable and secure device management and monitoring.

Projects

- **Real-Time Weather Data Monitoring & Forecasting System (Streaming ETL, AWS, Time-Series Analysis)**: Built a real-time ETL pipeline ingesting weather data from multiple APIs using AWS services for time-series forecasting and trend analysis. **Tech:** Python, AWS (S3, Lambda, EC2), PostgreSQL. [GitHub]
- **Real-Time Log Data Aggregation and Analytics (Cloud Data Engineering, Streaming Analytics)**: Engineered a scalable real-time log ingestion and processing pipeline on AWS to aggregate semi-structured application logs, transform them into structured formats, and enable live analytics dashboards. Designed schema optimization for historical analysis and performance monitoring. **Tech:** AWS (Kinesis/S3/EC2), Python, SQL, Snowflake [GitHub]
- **Automated COVID-19 Diagnosis from CT Scans (Machine Learning, Computer Vision)**: Developed a deep learning-based image classification model to detect COVID-19 from CT scan images, improving diagnostic efficiency through automated medical image analysis. Applied preprocessing, augmentation, and model evaluation techniques to enhance prediction accuracy. **Tech:** Python, TensorFlow/Keras, OpenCV.

Leadership & Volunteer Experience

- **Member – IEEE Student Chapter**: Organized and supported technical workshops and IEEE-led events, enhancing peer learning and professional development initiatives.
- **President – Student Activity Club (SAC)**: Spearheaded 5+ hackathons and campus-wide cultural events involving 300+ participants.